
MedFactCheck: Annotation Guidelines

1 Background

We have 5,755 atomic claims from 276 medical QA entries. Your job is to **group these atoms into snippets** and **label each snippet** with dual labels.

Each entry shows:

- **Query:** the medical question.
- **Full Response:** the LLM-generated answer (collapsible).
- **Atomic Claims:** individual statements extracted from the response, each with an expert label (✓ true / × false).

Why we merge at all (the motivation)

Atomic-claim extraction is often **too fine-grained**: a single atom like “*Option B is the correct answer*” or “*The factors include the individual’s age*” is **unverifiable in isolation**, because it needs the surrounding reasoning to make sense. Medical responses, especially clinical vignettes, chain claims together with *because / since / therefore / requires*. Splitting those chains destroys the verifiable unit.

We merge atoms whose logical dependency would be lost if each stood alone. Every rule below, and every judgment call during consensus, should trace back to this one question:

“If I read only this atom (or only this snippet), do I have enough to check whether it’s medically correct?”

If **yes** → atomic is fine. If **no** → merge with the atoms that carry the missing context.

2 Your Task

For each entry:

1. **Define Shared Context:** key-value pairs describing the query/response context (e.g., patient demographics, medications, topic). Click “Generate” to auto-extract, then edit as needed.
2. **Select atoms:** check the atoms in the left panel that belong together.
3. **Create Snippet:** click “+ Create Snippet” to group them.
4. **Generate or write text:** click “Generate” to get an LLM draft, or write the merged text yourself. Edit as needed.
5. **Set dual labels**, for each snippet:
 - **With context:** Is this correct *for this specific patient/query*?
 - **In general:** Is this correct as a *general medical statement*?
6. **Mark ambiguity:** check “Ambiguous” if you genuinely can’t determine correctness.
7. **Add notes:** explain your reasoning, especially when the two labels differ.
8. **Repeat** until all atoms are assigned to snippets.

3 Shared Context

Shared context is defined **per entry** and applies to all snippets. It captures the key facts from the query/response that snippets may reference.

For clinical vignettes:

Key	Example
age	23
sex	male
chief_complaint	shouting on subway, claiming little people trying to kill him
medical_history	marijuana and IV drug use, multiple suicide attempts
medications	haloperidol, diphenhydramine
vitals	temperature 98.6°F, BP 115/71, pulse 72
treatment_history	failure of risperidone, haloperidol, ziprasidone
correct_answer	Clozapine (B)

For consumer health queries:

Key	Example
topic	drug interaction
drug_a	NyQuil
drug_b	Benadryl

Click **“Generate”** to auto-extract context from the query. Review and edit: delete irrelevant keys, and fix values.

4 Grouping Rules (Merge vs Keep Atomic)

Guiding question (repeat it for every atom): *If this atom stood alone, could I still verify it?*

- **Yes, standalone is verifiable** → keep atomic.
- **No, it needs neighboring atoms for meaning** → merge.

4.1 Three patterns that trigger MERGE

All three were observed to hit full or near-full agreement in the test batch (6/6 or 5/6 annotators merged). If you see any of these, merge.

Pattern A: Enumeration of properties of one subject. The response lists multiple features, factors, or properties of the same thing. Merge the whole list into one snippet, even if each item is independently verifiable; the enumeration is the unit the LLM produced and the unit a reader evaluates.

Example A.1 (Entry 17, consumer query: “What is the most safe amount of Advil to take at one time?”).

atom 1: “The safe amount of Advil (ibuprofen) to take at one time depends on several factors.”

atom 2: “The factors include the individual’s age.”

atom 3: “The factors include the individual’s weight.”

atom 4: “The factors include the individual’s overall health.”

Decision: MERGE (6/6 annotators).

Why: atoms 2–4 begin with “The factors include...” and are meaningless without atom 1. The enumeration is the verifiable unit.

Example A.2 (Entry 75, vignette on a 72-year-old man with suspected leukemia).

atom 10: “Chronic Lymphocytic Leukemia (CLL) typically presents with a high leukocyte count.”

atom 11: “CLL typically presents with anemia.”

atom 12: “CLL typically presents with thrombocytopenia.”

Decision: MERGE (6/6 annotators).

Why: the response lists three clinical features that together form the “CLL picture”. The same pattern applies to atoms 13–15 (CML) and 16–18 (HCL); each leukemia subtype gets its own merged snippet.

Pattern B: Causal or conditional chain. Atoms connected by *because / since / as / due to / requires / therefore*, or where one atom is the premise for another. Merge so the logical link survives the decomposition.

Example B.1 (Entry 37, 23-year-old man with treatment-resistant schizophrenia in the ED).

atom 7: “Clozapine is generally reserved for patients who have failed to respond to at least two previous adequate trials of antipsychotic medications.”

atom 8: “Patients must have failed at least one conventional antipsychotic.”

atom 9: “Patients must have failed at least one second-generation antipsychotic.”

Decision: MERGE (6/6 annotators).

Why: atom 7 states the condition; atoms 8 and 9 specify the two required failures. Splitting would leave atoms 8 and 9 referring to an unstated requirement.

Example B.2 (Entry 9, consumer query on Hepatitis A IgM).

atom 7: “A non-reactive result does not necessarily rule out a past infection.”

atom 8: “IgM antibodies may not be detectable after a certain period.”

Decision: MERGE (5/6 annotators).

Why: atom 8 is the *reason* for atom 7 (non-reactive doesn’t rule out past infection *because* IgM fades). The causal link is the verifiable unit.

Pattern C: Conclusion + its supporting premises. A recommendation, diagnosis, or final answer that is only meaningful when read with the reasoning that produced it. Merge the conclusion with its premises.

Example C.1 (Entry 37, same schizophrenia vignette, end of the response).

atom 19: “The most appropriate next step in management is to initiate clozapine.”

atom 20: “Option B is the correct answer.”

Decision: MERGE (5/6 annotators).

Why: atom 20 is unverifiable alone; *what* is Option B? Atom 19 provides the referent.

Example C.2 (Entry 37, same vignette, earlier in the response).

atom 1: “The patient’s aggressive behavior is likely secondary to schizophrenia.”

atom 2: “The patient’s aggressive behavior could be due to a psychotic disorder.”

atom 3: “The most appropriate next step in management is to consider an alternative antipsychotic medication.”

Decision: MERGE (5/6 annotators).

Why: atoms 1–2 give the differential; atom 3 is the conclusion drawn from that differential.

4.2 Three patterns that justify KEEPING ATOMIC

Pattern D: Complete standalone fact, definition, or lab interpretation. The atom reads as a self-contained medical statement. No surrounding text needed.

Example D.1 (Entry 9 / atom 1, Hepatitis A IgM definition).

atom 1: “A non-reactive result for the Hepatitis A IgM (Immunoglobulin M) antibody test indicates that the individual does not have a recent or current Hepatitis A infection.”

Decision: ATOMIC (6/6 annotators).

Why: a complete definition of what a non-reactive IgM result means. Nothing else from the response is needed to judge it.

Example D.2 (Entry 75 / atom 1, lab value interpretation).

atom 1: “A platelet count of 119,000/mm³ is within normal limits.”

Decision: ATOMIC (6/6 annotators).

Why: a single lab-value claim verifiable in isolation. It happens to be **false** (the normal range is 150,000–400,000), but that is a labeling question, not a grouping one.

Pattern E: Topic genuinely shifts. The response moves from one subject to an unrelated one (e.g., drug composition → overdose risk, or one differential diagnosis → a different one). Split at the topic boundary. In Entry 75, the five leukemia subtypes (ALL, AML, CLL, CML, HCL) each got their own snippet, so **never merge across subjects**.

Pattern F: Same topic, but each atom carries a different checkable fact. Two claims about the same drug can still be separate if they convey distinct information. A standalone fact does not need to be dragged into a nearby enumeration just because it sits next to one.

4.3 When you’re torn

Default to **atomic**. It is cheaper and easier to merge during consensus review than to split a bad merge. But if Pattern A, B, or C clearly fires, merge without hesitation.

4.4 Consensus decisions from the test batch (apply these going forward)

These were real disagreements in our test batch. For each case below, the decision preserves the merge motivation. Use them as tie-breakers in the remaining batches.

Case 1: Entry 9 / atoms 9–10.

atom 9: “To confirm the status of Hepatitis A immunity, a Hepatitis A IgG (Immunoglobulin G) antibody test may be performed.”

atom 10: “A Hepatitis A IgG antibody test can detect antibodies that provide long-term immunity.”

Decision: MERGE.

Why: atom 10 specifies *what* the test in atom 9 does. Pattern B (extension). Without the merge, atom 9 refers to an unnamed test and atom 10 to an orphan test type.

Case 2: Entry 17 / atoms 5–6.

atom 5: “For adults and children over 12 years old, the typical over-the-counter dose is 200–400 mg every 4–6 hours as needed for pain or fever.”

atom 6: “The maximum daily dose for adults and children over 12 years old should not exceed 1200 mg unless directed by a healthcare provider.”

Decision: MERGE.

Why: both atoms describe the adult/adolescent dosing rule (Pattern A). A dosing recommendation needs both the per-dose amount and the daily cap to be useful.

Case 3: Entry 37 / atoms 4–6.

atom 4: “Clozapine is a second-generation antipsychotic.”

atom 5: “Clozapine is often used in treatment-resistant schizophrenia.”

atom 6: “Clozapine has a lower risk of extrapyramidal side effects.”

Decision: MERGE.

Why: enumerated properties of one drug: class, indication, side-effect profile (Pattern A). The majority (4/6) already merged.

Case 4: Entry 37 / atoms 10–12.

atom 10: “Clozapine can cause agranulocytosis.”

atom 11: “Agranulocytosis is a potentially life-threatening condition.”

atom 12: “Clozapine requires regular monitoring of the white blood cell count.”

Decision: MERGE all three.

Why: drug → side effect → required monitoring is a causal chain (Pattern B). Atom 12 alone (“Clozapine requires monitoring”) loses *what for* without atoms 10–11.

Case 5: Entry 37 / atoms 13–18.

atom 13: “Clozapine is initiated at a dose of 12.5–25 mg twice daily.”

atom 14: “Clozapine requires gradual titration to a therapeutic dose.”

atom 15: “The therapeutic dose of clozapine is typically in the range of 300–450 mg per day.”

atom 16: “Initiation of clozapine requires a trial of 2 weeks.”

atom 17: “During the trial period, the patient is monitored for agranulocytosis.”

atom 18: “During the trial period, the patient is monitored for other potential side effects.”

Decision: Two snippets: {13, 14, 15} as the dosing protocol, and {16, 17, 18} as the trial-and-monitoring protocol.

Why: atoms 13–15 describe the dose regimen; atoms 16–18 describe the trial duration and what is being monitored. Pattern A applies within each group. Merging all six would create a long blob spanning two different sub-topics.

Case 6: Entry 75 / atoms 19–25.

atom 19: “Based on the patient’s age, symptoms, and laboratory results, the most likely diagnosis is Chronic Lymphocytic Leukemia (CLL).”

atom 20: “Chronic Lymphocytic Leukemia (CLL) is a type of cancer that affects the blood and bone marrow.”

atom 21: “CLL is characterized by the production of an excessive number of immature white blood cells (lymphocytes).”

atom 22: “CLL is most commonly diagnosed in individuals over the age of 50.”

atom 23: “CLL often presents with a high leukocyte count.”

atom 24: “CLL often presents with anemia.”

atom 25: “CLL often presents with a mild enlargement of the spleen.”

Decision: Two snippets: {19} alone (the final diagnosis), and {20–25} as the CLL profile expansion.
Why: atom 19 is the answer to the case; atoms 20–25 are the LLM expanding on what CLL is. Avoid the 7-atom blob that 3 annotators produced.

Every atom must be assigned. The dashboard shows an “unassigned” counter. The Next button warns you if atoms are left unassigned.

5 Snippet Text

The snippet text should be a **self-contained verifiable statement**. A reader should be able to judge its correctness without reading the original query.

Click “Generate” to get an LLM draft. The LLM will:

- Merge the selected atoms into coherent text.
- Resolve pronouns using the shared context (e.g., “the patient” → “a 23-year-old male”).
- Keep standalone facts as-is (no unnecessary context added).
- NOT add medical facts not present in the atoms.

Always review and edit the generated text. The LLM is a starting point, not final.

6 Dual Labels

Each snippet gets TWO labels.

What “context” refers to: the information in the user **query**, plus the **earlier statements in the same response** that precede the current snippet. It is *not* outside medical knowledge or anything from other entries.

Label	Question	Example (Entry 37)
In general	Is this correct as a <i>general medical statement</i> ?	“Initiation of clozapine requires a 2-week trial” → True (a short initial trial is reasonable in general).
With context	Is this correct <i>for this specific patient/query</i> ?	Same snippet → False (this treatment-resistant patient needs a longer trial).

When the two labels differ, flag it! This is the most interesting case for our research: “**generally true, contextually wrong**” (or vice versa). Always add a note explaining why.

Label values.

- **True:** the snippet is medically/factually correct.
- **False:** the snippet contains a factual error.
- If unsure, mark **Ambiguous** and add a note.

7 Worked Examples (all from the test batch)

These four labeling cases come directly from the test batch. They cover the four label configurations you will encounter: both-true, both-false (medical error), context-dependent split, and single-atom lab error.

Example 1: Consumer, both labels True

Entry 17 (consumer query): “What is the most safe amount of Advil to take at one time?”

Source atoms (1–4):

atom 1: “The safe amount of Advil (ibuprofen) to take at one time depends on several factors.”

atom 2: “The factors include the individual’s age.”

atom 3: “The factors include the individual’s weight.”

atom 4: “The factors include the individual’s overall health.”

Merged snippet:

“The safe amount of Advil (ibuprofen) depends on several factors, including age, weight, and overall health.”

Labels: in general = **True**; with context = **True**.

Grouping: Pattern A (enumeration). All 6 annotators merged.

Reasoning: the statement is generically correct and nothing in a consumer query changes it.

Example 2: Consumer, both labels False (medical error)

Entry 9 (consumer query): “So what does the non reactive mean for the hep a igm”

Source atoms (4–6):

atom 4: “A non-reactive result suggests that the individual may have been vaccinated against Hepatitis A.”

atom 5: “A non-reactive result suggests that the individual has a past infection that is no longer active.”

atom 6: “A non-reactive result suggests that the individual has never been infected with the Hepatitis A virus.”

Merged snippet:

“A non-reactive Hep A IgM result suggests that the individual may have been vaccinated against Hepatitis A, has a past infection that is no longer active, or has never been infected with the Hepatitis A virus.”

Labels: in general = **False**; with context = **False**.

Grouping: Pattern A (enumeration of three interpretations of “non-reactive”).

Reasoning: vaccination and past infection are detected by **IgG**, not IgM; IgM detects recent or current infection. Both labels are False because the fact itself is wrong; no context can rescue it.

Example 3: Vignette, context-dependent (the key case for our paper)

Entry 37 (vignette): “A 23-year-old man in the ED, treatment-resistant schizophrenia, failed risperidone, haloperidol, and ziprasidone...”

Source atoms (16–18):

atom 16: “Initiation of clozapine requires a trial of 2 weeks.”

atom 17: “During the trial period, the patient is monitored for agranulocytosis.”

atom 18: “During the trial period, the patient is monitored for other potential side effects.”

Merged snippet:

“Initiation of clozapine requires a trial of 2 weeks, during which the patient is monitored for agranulocytosis and other potential side effects.”

Labels: in general = **True**; with context = **False**.

Note (required when labels differ): in general, a short initial observation period is reasonable for starting clozapine. **For this specific treatment-resistant patient**, the expected trial period before declaring treatment failure is longer than 2 weeks, so the “2 weeks” claim is contextually wrong.

Why this example matters: this is the failure mode our paper targets, namely *generally plausible, contextually wrong*. When the two labels diverge, a note is mandatory; it is the artifact our analysis depends on.

Example 4: Vignette, single-atom lab error

Entry 75 (vignette): “A 72-year-old man, fatigue, leukocyte count 67,500, platelet count 119,000...”

Source atom (1):

atom 1: “A platelet count of 119,000/mm³ is within normal limits.”

Snippet (kept atomic, same text):

“A platelet count of 119,000/mm³ is within normal limits.”

Labels: in general = **False**; with context = **False**.

Grouping: Pattern D (standalone lab interpretation). All 6 annotators kept atomic.

Reasoning: 119,000/mm³ is below the normal range (150,000–400,000). Both labels are False: a pure medical error.

Quick reference: when to write a note

- **Always** when *in general* $\neq$ *with context*.
- **Always** when you mark a snippet Ambiguous.
- **Optional** when the medical error is subtle, or when you made a non-obvious grouping choice.

8 Suggestions (Optional)

The dashboard offers two optional features using the pre-clustered LLM groupings:

- **“Show suggestions”** checkbox (left panel): overlays purple borders on atoms showing the LLM’s original grouping. Off by default to avoid bias.
- **“Import Suggestions”** button: loads pre-clustered snippets as editable cards. Use if you agree with the grouping, but you must still set dual labels.

Both are optional. You can ignore them and cluster entirely from scratch.

9 Dashboard Workflow

1. Open your assigned batch from the home screen.
2. For each entry:
 - Read the **query** (and optionally the full response).
 - Click **“Generate”** on Shared Context to auto-extract, then review.
 - Select atoms by clicking checkboxes → **“+ Create Snippet”**.
 - Click **“Generate”** on each snippet for draft text → edit as needed.
 - Set **both labels** (With context + In general) and mark ambiguity if needed.
 - Add **notes** when labels differ or when something is interesting.
 - Click **Next** → (warns if anything is incomplete).
3. Click **“Export Batch”** to download your annotations as JSON.
4. Export regularly as backup.

Keyboard shortcuts: ← → to navigate entries, Escape to go back to batches.

10 Batch Assignments

Annotator	Batches
Annotator 1	1, 7, 13, 19, 25
Annotator 2	4, 10, 16, 22
Annotator 3	6, 12, 18, 24, 28
Annotator 4	2, 11, 17, 23, 27
Annotator 5	5, 8, 14, 20, 26
Annotator 6	3, 9, 15, 21

Start with the **Test Batch** for practice.

Dashboard: <https://mbzuai-nlp.github.io/medfactcheck/>